

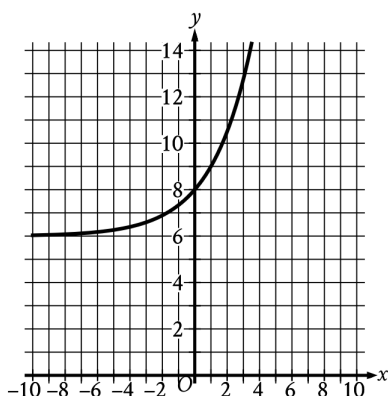
SAT UZBEKISTAN

Telegram @satuzbhunters

Module-1

1

Mark for Review



(A) $(-8, 0)$

(B) $(-6, 0)$

(C) $(0, 6)$

(D) $(0, 8)$

2

Mark for Review



The function f is defined by $f(x) = -3x + 60$. What is the value of $f(x)$ when $x = -8$?

(A) 49

(B) 52

(C) 57

(D) 84

A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

(A) $4xy = 70$

(B) $4(x + y) = 70$

(C) $3x + y = 70$

(D) $x + 3y = 70$

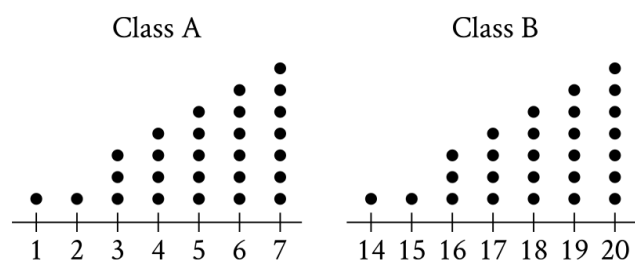
A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

(A) 3

(B) 7

(C) 36

(D) 44



Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

(A) The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.

(B) The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.

(C) The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.

(D) There is not enough information to compare these standard deviations.

6

 Mark for Review

Which expression is equivalent to $(7x^3 + 7x) - (6x^3 - 3x)$?

(A) $x^3 + 10x$

(B) $-13x^3 + 10x$

(C) $-13x^3 + 4x$

(D) $x^3 + 4x$

7

 Mark for Review

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

(A) 36

(B) 45

(C) 72

(D) 108

8

 Mark for Review

$$3x(x - 4)(x + 5) = 0$$

What is one of the solutions to the given equation?

(A) -4

(B) 0

(C) 3

(D) 5

9

 Mark for Review

A chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL). The volume of acetic acid in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

10

 Mark for Review

A distance of 354 furlongs is equivalent to how many feet?

(1 furlong = 220 yards and 1 yard = 3 feet)

(A) 306

(B) 402

(C) 25,960

(D) 233,640

11

 Mark for Review

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the xy -plane?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

12

 Mark for Review

13 is $p\%$ of 25. What is the value of p ?

13

 Mark for Review

The function $f(t) = 60,000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

14

 Mark for Review


Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

15

 Mark for Review

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

16

 Mark for Review


In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

(A) $\frac{\sqrt{15}}{15}$

(B) $\frac{\sqrt{15}}{4}$

(C) $\frac{4\sqrt{15}}{15}$

(D) $\sqrt{15}$

17

 Mark for Review


$$p = \frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

(A) $4j + 9 = \frac{k}{p}$

(B) $4j + 9 = kp$

(C) $4j + 9 = k - p$

(D) $4j + 9 = \frac{p}{k}$

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

(A)

x	y
3	21
5	47
8	86

(B)

x	y
3	26
5	42
8	86

(C)

x	y
3	16
5	42
8	81

(D)

x	y
3	26
5	52
8	91

Line ℓ is defined by $3y + 12x = 5$. Line n is perpendicular to line ℓ in the xy -plane. What is the slope of line n ?

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy -plane, how many distinct x -intercepts does the graph have?

(A) 4

(B) 3

(C) 2

(D) 1

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

(A) $f(x) = 50(1.6)^{x+1}$

(B) $f(x) = 80(1.6)^x$

(C) $f(x) = 128(1.6)^{x-1}$

(D) $f(x) = 204.8(1.6)^{x-2}$

$$f(x) = 4x^2 + 64x + 262$$

The function g is defined by $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

(A) -13

(B) -8

(C) -5

(D) -3

SAT UZBEKISTAN

Telegram @satuzbhunters

Module-2

1

 Mark for Review



$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

(A) $4x = 108$

(B) $4x = 24$

(C) $4x = 12$

(D) $4x = 3$

2

 Mark for Review



The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is \$400?

(A) \$13,400

(B) \$13,000

(C) \$15,400

(D) \$37,400

3

 Mark for Review



If the graph of $27x + 33y = 297$ is shifted down 5 units in the xy -plane, what is the y -intercept of the resulting graph?

(A) (0, 4)

(B) (0, 6)

(C) (0, 14)

(D) (0, 28)

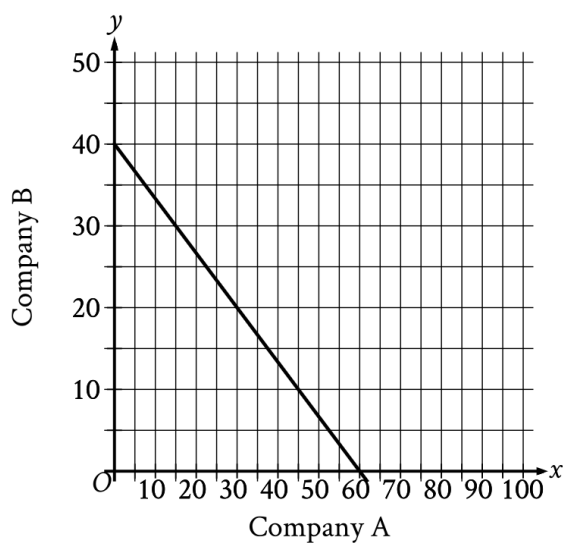
The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

(A) $(-2, 0)$

(B) $(0, -2)$

(C) $(0, \frac{1}{10})$

(D) $(\frac{1}{10}, 0)$



The graph shows the relationship between the number of shares of stock from Company A, x , and the number of shares of stock from Company B, y , that Simone can purchase. Which equation could represent this relationship?

(A) $y = 8x + 12$

(B) $8x + 12y = 480$

(C) $y = 12x + 8$

(D) $12x + 8y = 480$

$$\frac{7}{2}x + 6y = 25$$

$$\frac{5}{2}x + 6y = 23$$

The solution to the given system of equations is (x, y) . What is the value of $\frac{17}{2}x + 18y$?

(A) 2

(B) 3

(C) 48

(D) 71

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

(A)

x	y
-6	-9
-5	-8
4	1

(B)

x	y
-6	-3
-5	-3
4	-3

(C)

x	y
-6	-3
-5	-2
4	7

(D)

x	y
-6	3
-5	3
4	3

8

 Mark for Review


For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

(A) $q(x) = 0.55(14)^x$

(B) $q(x) = 1.45(14)^x$

(C) $q(x) = 14(0.55)^x$

(D) $q(x) = 14(1.45)^x$

9

 Mark for Review

A submersible device is used for ocean research. The function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

10

 Mark for Review


A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

(A) 149,796

(B) 164,500

(C) 190,955

(D) 310,800

11

 Mark for Review

$$x(x + 1) - 56 = 4x(x - 7)$$

What is the sum of the solutions to the given equation?



$$y < 6x + 2$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

(A)

x	y
3	20
5	32
7	44

(B)

x	y
3	16
5	36
7	40

(C)

x	y
3	16
5	28
7	40

(D)

x	y
3	24
5	36
7	48



What is the diameter of the circle in the xy -plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

(A)

4

(B)

8

(C)

16

(D)

32

$$-9x^2 + 30x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

(A) 3

(B) 0

(C) -25

(D) -53

In triangle XYZ , angle Y is a right angle, the measure of angle Z is 33° , and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant, what is the value of k ?

The regular price of a shirt at a store is \$11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the \$ sign when entering your answer. For example, if your answer is \$4.97, enter 4.97)

In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

(A) The measure of angle A

(B) The length of side AB

(C) The lengths of sides BC and EF

(D) No additional information is necessary.

18

 Mark for Review

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

19

 Mark for Review

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

(A) $3x^2 + 7x + 14b$

(B) $3x^2 + 28x + 14b$

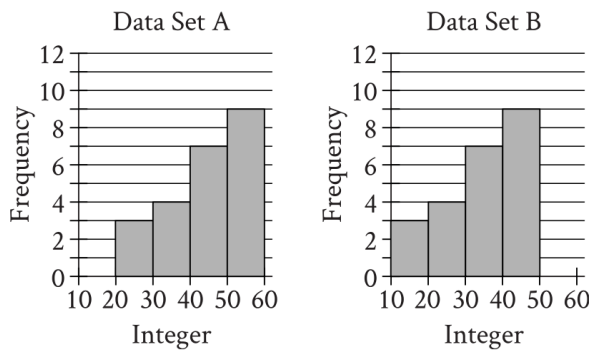
(C) $3x^2 + 42x + 14b$

(D) $3x^2 + 49x + 14b$

20

 Mark for Review

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?



Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- (A) 0
- (B) 1
- (C) 10
- (D) 23

Poll Results	
Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

- (A) 163
- (B) 1,304
- (C) 3,864
- (D) 5,621